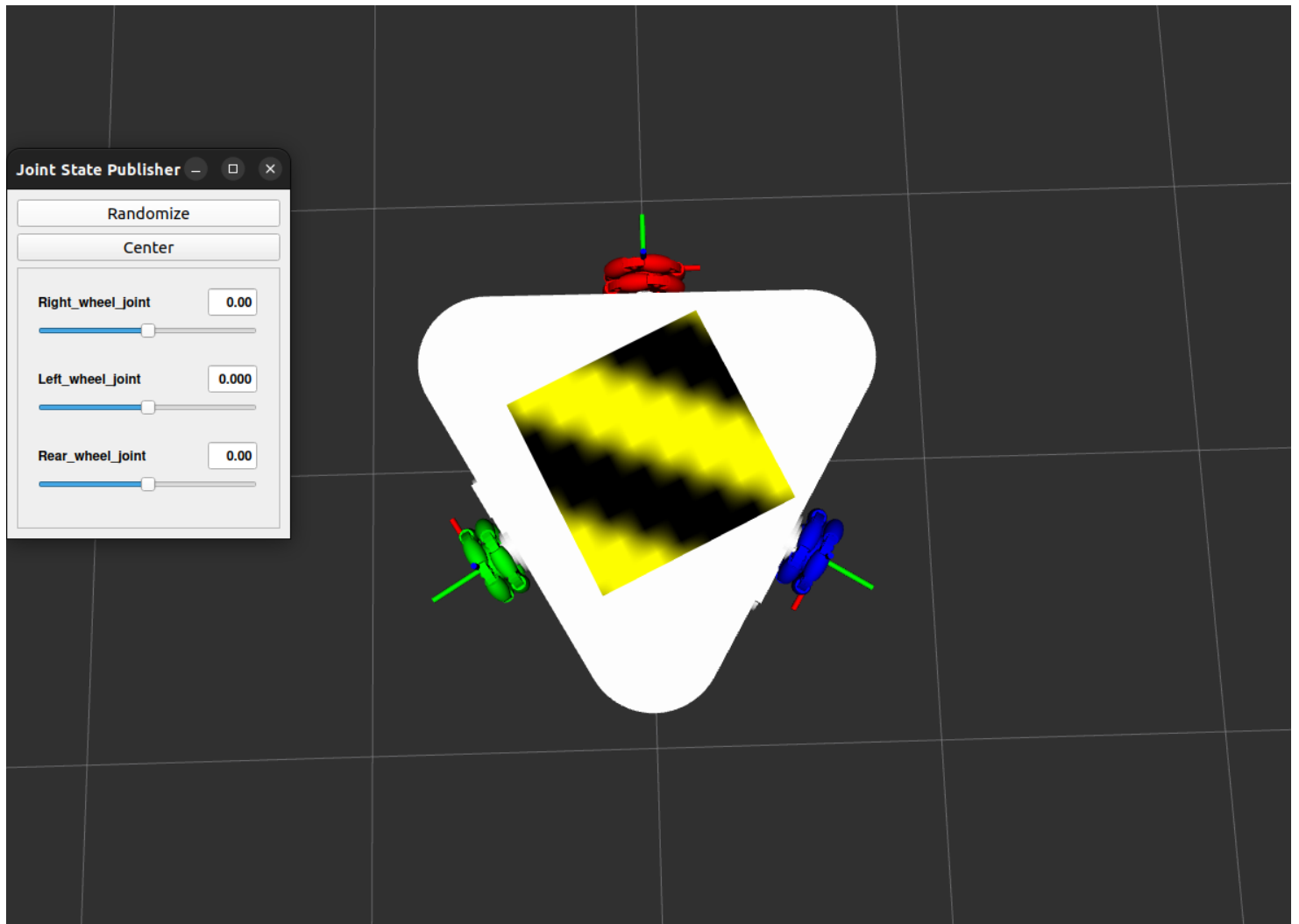


Kinematics of bot



In the above bot the left wheel
i.e. red, right wheel - blue,
mean wheel - green.

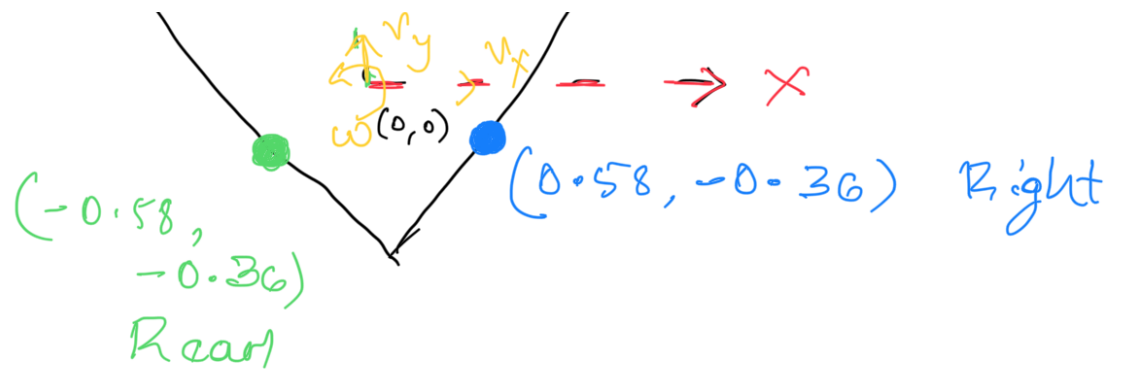
If you check the world you
will get their coordinates defined
in terms of baseline.

Our job is to derive the forward
kinematics & inverse kinematics of
the bot.

Inverse Kinematics

→ we have bot's linear
and angular velocity and
wish to find the corre-
sponding wheel angular
velocities.



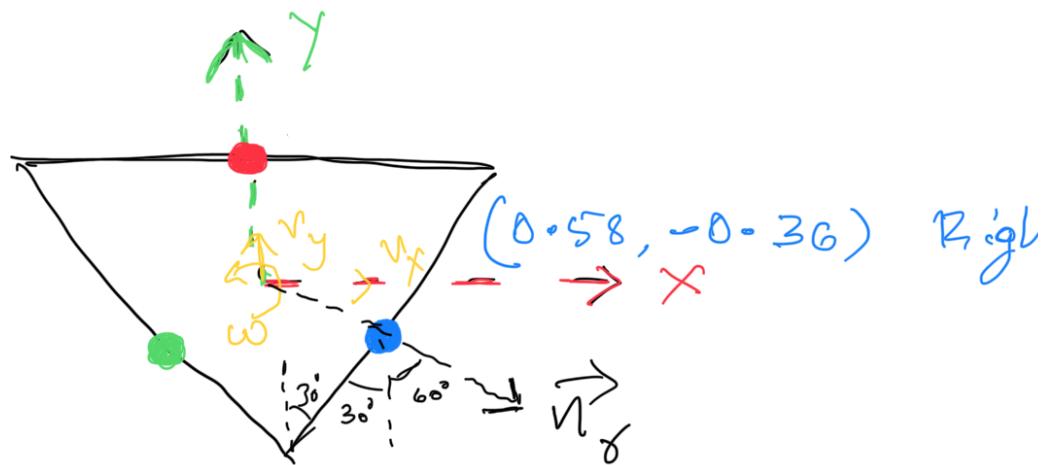


Let the bot is moving with linear velocity v_x & v_y and angular velocity w .

$$\text{So } \vec{v}_{bot} = \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix} \text{ \& } \vec{\omega}_{bot} = \begin{bmatrix} 0 \\ 0 \\ w \end{bmatrix}$$

Now we just have to calculate each velocities about each of the wheels.

I am doing for the right, you may follow the same procedure for rest wheels.



$\vec{n}_x$ will be a normal vector for right wheel.

$$\text{So } \vec{n}_{\text{right}} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix}$$

Dist. between right & centre =

$$\sqrt{(0.58-0)^2 + (-0.36-0)^2}$$

$$\Rightarrow r \approx 0.68$$

$$\text{So } \vec{r}_{\text{right}} = r \vec{n}_r = \begin{bmatrix} 0.58 \\ -0.36 \\ 0 \end{bmatrix}$$

$$\text{Now } \vec{v}_{\omega \text{ right}} = \vec{\omega} \times \vec{r}_{\text{right}}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix} \times \begin{bmatrix} 0.58 \\ -0.36 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.34\omega \\ 0.58\omega \\ 0 \end{bmatrix}$$

Got the angular part now translational part.

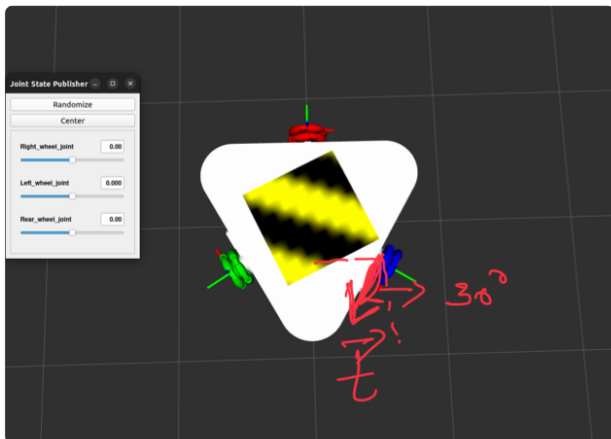
$$\vec{v}_{\text{Tright}} = \vec{v}_{\text{body}} = \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix}$$

$$\text{So } \vec{v}_{\text{total right}} = \vec{v}_T + \vec{v}_\omega$$

$${}^{g_1}z \begin{bmatrix} v_x + 0.34w \\ v_y + 0.58w \\ 0 \end{bmatrix}$$

** The right wheel can only move radially, not in any other direction. So we need that translation component of $\vec{v}_{total}$.

$\Rightarrow$ To determine open viz & increment the joint angle of right wheel in the direction & observe along which the wheel is rotating.



I observed the right wheel about that $\vec{t}$ vector dirn.

$$\vec{t} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}$$

$$|\vec{v}_{right}| = \vec{t} \cdot \vec{v}_{total}$$

$$= \begin{bmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} v_x + 0.34w \\ v_y + 0.58w \\ 0 \end{bmatrix}$$

$$|\vec{\omega}_{\text{right}}| = \frac{v_{\text{right}}}{r_{\text{right}}} = \frac{-\frac{v_x}{2} - \frac{\sqrt{3}}{2} v_y - 0.68\omega}{\text{radius of wheel}}$$

Similarly implement for other two wheels.

You will get a different PK equation if you take a different coordinate system or procedure. If your result is correct, then motion should also be correct.

You can verify PK equation for checking motion for translational, lateral & angular motion individually.

Forward Kinematics $\rightarrow$ we have angular velocities of wheels & want to determine bot's velocity.

From IK you will get $\omega_{right} = A v_x + B v_y + C \omega$

$$\omega_{left} = \dots$$

$$\omega_{mean} = \dots$$

Express v_x, v_y, ω in terms of $\omega_{right}, \omega_{left}, \omega_{mean}$. That's it; you will get FK eqn.